

RAMESH RANGARAO PATIL

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Summary

Backend and ML Engineering professional with 2+ years of experience building production-grade ML systems, GenAI infrastructure, and scalable microservices backends. Proficient in Python, FastAPI, and PyTorch with expertise in ML inference, vector search, RAG architectures, agentic orchestration, and ML observability. Experienced in taking end-to-end ownership — from researching third-party integrations and designing ML pipelines to production deployment, drift monitoring, and automated retraining.

Experience

Dynamisch IT Pvt. Ltd.

Software Engineer

Jan 2024 – Present

Pune, Maharashtra

Event-Driven AI Orchestration & Graph RAG Pipeline — Investigation App

- Designed a distributed, event-driven pipeline (FastAPI, RabbitMQ, Celery) to ingest multi-modal files (video, audio, documents) and orchestrate multiple ML models to process them into structured Markdown artifacts.
- Built end-to-end ingestion into a knowledge graph with Graph RAG-based retrieval; implemented production observability with latency and error-rate alerting across the pipeline.

Audio Fingerprinting & ML Inference Service — Wazzdat (Microservices Mobile App)

- Built the core ML microservice in FastAPI for real-time audio fingerprinting and music identification; designed and trained a custom CNN-based 128-D embedding model, owning the full ML lifecycle from data preparation to deployment.
- Optimized large-scale vector search using Milvus (IVF/HNSW indexing) with Redis caching, achieving sub-100ms retrieval latency; implemented request-level caching and routing to reduce redundant inference calls.

ML Observability & Automated Retraining Service — Wazzdat

- Built a standalone monitoring microservice to detect embedding drift and retrieval quality degradation; automated model retraining and Milvus index retuning triggered by drift signals with zero manual intervention.
- Developed evaluation pipelines to benchmark retrieval accuracy against gold-standard datasets, establishing measurable release criteria before each production rollout.

Concert Events Platform — Wazzdat (Golang Admin & User Services)

- Took end-to-end ownership of the concert events feature: researched affiliate programs, onboarded multi-platform provider APIs, and built Golang admin service modules with cron schedulers for continuous, fault-tolerant data ingestion.
- Implemented user service modules for concert discovery — geospatial filtering, city/genre-wise browsing, nearby events, trending algorithms, ticketing, and user interactions — with Redis caching reducing API response times by 40%.

Healthcare RAG Chatbot & Data Visualization Platform

- Built a RAG-based chatbot using LangChain and Google Gemini API with document ingestion, chunking, embedding, and vector retrieval pipelines; applied prompt engineering strategies to reduce hallucination rates.
- Developed a secure backend platform with Stripe payment integration, OTP and Two-Factor Authentication (2FA), and data visualization dashboards.

Personal Projects

Resume ATS Scorer & Optimizer | Python, LLMs, OCR, Evaluation Pipelines

- Agentic AI system using LLMs to score job-fit alignment, benchmark ATS parsing quality, and generate structured candidate assessment reports with cross-version evaluation metrics.

Agentic AI Suite | Python, Agno, RAG, LLMs, Agentic Workflows

- Autonomous AI agents with tool-using capabilities (RAG chatbot, PDF summarizer, email generator); multi-step orchestration with safety guardrails and structured output validation.

Computer Vision & Geospatial Segmentation | Python, PyTorch, U-Net, YOLOv8

- Aerial land cover segmentation (U-Net) for roads, urban areas, and water bodies; YOLOv8 number plate detector trained on 2,000+ annotated images with integrated OCR extraction.

Technical Skills

Languages: Python, Golang, C, C++

ML & AI Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, Scikit-learn, ONNX

LLM & GenAI: ML Inference, RAG Architectures, Prompt Engineering, Agentic Orchestration, Google Gemini API

Backend & Infra: FastAPI, Django, Flask, Microservices, REST APIs, RabbitMQ, Celery, Docker, CI/CD

ML Ops & Observability: ML Monitoring, Drift Detection, Evaluation Pipelines, Automated Retraining, Alerting

Databases & Vector DB: Milvus, Elasticsearch, PostgreSQL, MySQL, MongoDB, Redis

Cloud & Tools: AWS (EC2, S3), NumPy, Pandas, Matplotlib, librosa, Git

Education

SKN Sinhgad Institute of Technology and Science

B.E. Computer Engineering (Honors in Robotics) | CGPA: 8.28

Dr. DY Patil School of Engineering (Polytechnic)

Diploma in Computer Engineering | 77.50%

Aug 2020 – July 2023

Pune, Maharashtra

July 2017 – Oct 2020

Pune, Maharashtra